# Chapter 1

# Introduction

## 1.1 Background

Metabolic dysfunction-associated steatotic liver disease (MASLD), formerly non-alcoholic fatty liver disease (NAFLD), is the most common chronic liver disease worldwide. It affects about 30% of the adult population, and its prevalence has risen steadily over the past three decades [1, 2]. In 2023 a multisociety consensus renamed the condition to reflect its metabolic origin: steatotic liver disease accompanied by at least one cardiometabolic risk factor, such as excess weight, raised blood glucose or dyslipidaemia [3]. Untreated, MASLD can progress from simple fat accumulation (steatosis) to inflammation, fibrosis and cirrhosis, and it is closely linked to type 2 diabetes and cardiovascular disease.

Diet is central to its management. No dietary pattern is a cure, but the current European guideline recommends lifestyle change, centred on diet quality and weight loss, as the first-line treatment for most patients [4]. The evidence is graded by the amount of weight lost: in a prospective study with paired liver biopsies, a loss of at least 5% of body weight reduced steatosis, 7–10% improved inflammation, and 10% or more led to resolution of steatohepatitis in most patients and to regression of fibrosis in many [5]. Beyond weight, the guideline advises a Mediterranean-style diet, fewer foods rich in saturated fat and sugars, and the avoidance of sugar-sweetened beverages [4].

Turning this guidance into daily meals is where patients struggle. Guidelines speak in nutrients and dietary patterns (“limit saturated fat”, “increase fibre”), whereas people decide what to eat one recipe at a time, constrained by the cuisine they know, the ingredients they have, the time they can spend cooking and what they enjoy. The same advice also does not suit every patient equally: the priority for someone whose diet is already high in fibre but rich in saturated fat differs from that of someone with the opposite pattern. Bridging the gap between population-level evidence and an individual patient’s plate is the problem this thesis addresses.

Liver fat itself can be measured non-invasively. Vibration-controlled transient elastography (FibroScan) reports the controlled attenuation parameter (CAP), which grades steatosis with established cut-offs [6]. The US National Health and Nutrition Examination Survey (NHANES) performed this examination on a nationally representative sample in its 2017–March 2020 and 2021–2023 cycles, together with blood tests, body measurements, questionnaires and a 24-hour dietary recall [7]. In the 15,300 participants with a complete examination used in this work, 52% had a CAP of at least 248 dB/m, the usual threshold for any steatosis.

## 1.2 Digital Twins in Medicine and Nutrition

A digital twin is a virtual representation of a specific physical system that is updated with data from that system, can predict how it will behave, and is used to inform decisions about it [8]. In medicine the physical system is a patient, or one of their organs. Medical digital twins are most advanced in cardiology and oncology, and their development faces common challenges in data integration, validation and privacy [9, 10]. What distinguishes a twin from a one-off prediction model is that it is specific to one patient, is updated as new data arrive, and can simulate how that patient's state may change under different interventions [8, 10].

Nutrition is a natural application. People respond differently to the same food [11], and the "virtual digital twin" concept in precision nutrition proposes combining a person's clinical, lifestyle and dietary data into a model that can test dietary choices before they are made [12]. Early clinical results are encouraging: in a one-year randomised trial in people with type 2 diabetes, nutrition guided by a digital twin that predicted each person's post-meal glucose response improved markers of liver fat [13]. In MASLD specifically, the European ARTEMIS project is building virtual twins from retrospective data on several thousand patients [14]. None of these systems, however, evaluates complete, existing recipes against a patient's evidence-based dietary needs.

## 1.3 Computational Gastronomy and Recipe Data

Computational gastronomy treats food and cooking as data that can be analysed and reasoned about [15]. Structured resources now make this possible at scale. RecipeDB contains more than 118,000 recipes from 74 countries, each with its ingredients, cooking processes, utensils, cuisine and nutritional profile [16]; FlavorDB describes the flavour molecules found in foods [17]; and literature-mined resources link culinary herbs and spices to health outcomes [18]. Related work in menu planning uses multi-objective optimisation to compose daily menus that meet nutritional targets [19].

These resources are designed mainly for exploring and querying culinary data, or for meeting generic nutrient targets. They do not, by themselves, decide which existing recipes suit a particular patient, given that patient's current liver state, the strength of the clinical evidence behind each dietary change, and the practical realities of their kitchen.

## 1.4 Problem Statement and Research Gap

Three bodies of work hold complementary pieces of the problem. Clinical evidence links dietary changes to liver outcomes; patient-specific models, including digital twins, estimate an individual's health state; and recipe databases describe the food people actually cook. These components are usually studied separately. In particular, we found no framework that evaluates and ranks existing recipes for an individual patient by combining

- **health relevance**, derived from the patient's current liver and metabolic state and from graded intervention evidence;
- **cuisine compatibility**, reflecting the patient's culinary context; and
- **practical feasibility**, reflecting cooking time, number of ingredients and complexity,

within a transparent framework in which every recommendation can be traced back to the evidence that supports it.

## 1.5 Research Question and Objectives

This thesis asks:

> *Can a patient-specific liver digital twin, built from clinical data and graded intervention evidence, support the evidence-grounded evaluation and ranking of existing recipes according to their health relevance, cuisine compatibility and cooking feasibility for an individual patient?*

To answer it, the work pursues five objectives:

1. Estimate each patient's current liver state, as calibrated probabilities of steatosis, from routinely collected clinical data.
2. Translate that state, together with the patient's own reported diet, into evidence-graded nutritional priorities, without inventing individual numeric targets.
3. Evaluate and rank existing recipes from RecipeDB against those priorities, the patient's cuisine and practical constraints.

4. Compare recipe choices as scenarios whose projected effects are grounded in clinical evidence, and update the patient's state as new data arrive.
5. Evaluate each component against appropriate baselines and state clearly what has and has not been validated.

## 1.6 Approach Overview

The proposed system, HepatoTwin, is organised in seven layers (Figure 1.1). Layer 1 holds the patient's state: demographics, body measurements, blood tests, questionnaire answers and a 24-hour dietary recall. Layer 2, the core of the twin, estimates the current liver state with an XGBoost model and two calibrated classifiers that give the probability of CAP at or above 248 and 280 dB/m, explained per feature with SHAP values. Layer 3, the evidence engine, converts this state into nutritional priorities such as weight management or limiting saturated fat; each priority cites rows of a curated table of 23 intervention–outcome findings with explicit evidence-strength labels, and is adjusted by the patient's own intake. Layer 4, the recipe evaluation and ranking engine (NutriTwin), is the core research contribution: it scores candidate recipes for health relevance, cuisine compatibility, practical feasibility and acceptability, ranks first the recipes that meet every evidence-linked reference value, and explains each recommendation. Layer 5 compares recipe scenarios and projects their effect over time; Layers 6 and 7 present the twin through a three-dimensional liver visualisation and an explainable dashboard. When new measurements or food logs arrive, the twin updates its state and its recommendations.

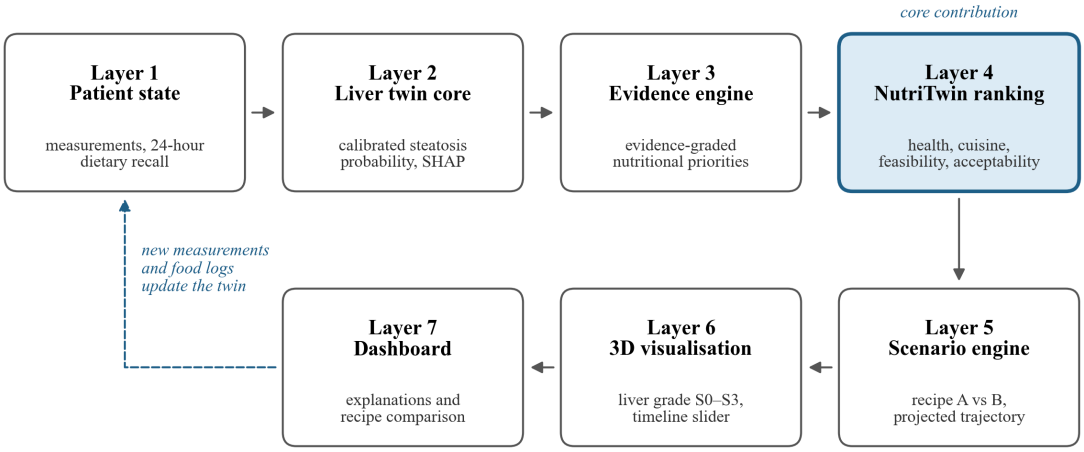



Figure 1.1: Overview of HepatoTwin. The patient's data (Layer 1) feed the liver twin core (Layer 2), whose calibrated state drives evidence-graded priorities (Layer 3) and the ranking of existing recipes (Layer 4, the core contribution). Scenarios (Layer 5) are shown through a visualisation (Layer 6) and a dashboard (Layer 7); new measurements and food logs update the twin.

The twin is built and evaluated on NHANES. Its liver-state model is intended for patients aged 12 to 59 and is evaluated against reliable FibroScan examinations; ages 60 and over are outside its scope.

## 1.7 Contributions

The main contributions of this thesis are:

1. **A calibrated liver-state model from routine data.** Trained on both NHANES FibroScan cycles, the model detects steatosis with ROC-AUC 0.881 (CAP ≥ 248 dB/m) and 0.879 (CAP ≥ 280 dB/m) on held-out patients within its scope, above the Fatty Liver Index (0.861) and the Hepatic Steatosis Index (0.836) on the same patients [20, 21], and holds its performance on a later survey cycle. It also predicts CAP with a held-out $R^2$ of 0.56, comparable to or above published models built on routine data.
2. **An evidence engine linking patient state to graded dietary priorities.** Every priority cites named trials, meta-analyses or guidelines with an explicit strength label, observational findings are recorded but never used as effects, and the patient's own 24-hour recall personalises the priorities.
3. **NutriTwin, an evidence-gated multi-criteria recipe ranking engine.** On 300 held-out patients, 83% of its top-ten recipes met every evidence-linked reference value, the patient's cuisine and the practical limits at once, against 73% for

weighted scoring without the health gate and under 2% for health-only ranking; it was the only method whose recommendations always met the nutritional references.

4. **Scenario comparison and weekly meal plans** that respect the patient's cuisine and nutritional references, with projected effects labelled by the strength of their evidence.
5. **An open, modular digital-twin framework**, demonstrated for MASLD, together with an in-silico evaluation design that tests how the twin updates on virtual patients before real follow-up data exist.

## 1.8 Scope and Limitations

NHANES is cross-sectional: each participant was examined once. The twin's estimate of the current state is therefore validated on held-out patients, but its projections of future states cannot be compared with real repeat measurements. They are presented as simulated projections grounded in clinical evidence, never as clinical forecasts, and the twin's updating is evaluated on virtual patients. Validation with real longitudinal data requires deployment with patients who enter their own measurements over time. The recommendations are research outputs: they support, and do not replace, clinical care.

## 1.9 Thesis Outline

Chapter 2 reviews related work on MASLD and its dietary treatment, non-invasive liver fat assessment, machine-learning prediction of steatosis, dietary assessment, digital twins, computational gastronomy and the evaluation of twins without longitudinal data. Chapter 3 describes the methodology: the NHANES cohort and the recipe corpus, the liver-state model, the evidence engine, the recipe evaluation and ranking engine, and the scenario and updating components of the twin. Chapter 4 reports the experiments carried out to improve the prediction of CAP, and explains why its performance reaches a plateau. Chapter 5 presents the results of each component against its baselines.

# References


1. Younossi, Z. M., Golabi, P., Paik, J. M., Henry, A., Van Dongen, C. and Henry, L. The global epidemiology of nonalcoholic fatty liver disease (NAFLD) and nonalcoholic steatohepatitis (NASH): a systematic review. *Hepatology*, 77(4):1335–1347, 2023. doi:10.1097/HEP.0000000000000004.
2. Riazi, K., Azhari, H., Charette, J. H., Underwood, F. E., King, J. A., Afshar, E. E., Swain, M. G., et al. The prevalence and incidence of NAFLD worldwide: a systematic review and meta-analysis. *The Lancet Gastroenterology & Hepatology*, 7(9):851–861, 2022. doi:10.1016/S2468-1253(22)00165-0.
3. Rinella, M. E., Lazarus, J. V., Ratziu, V., Francque, S. M., Sanyal, A. J., Kanwal, F., et al. A multisociety Delphi consensus statement on new fatty liver disease nomenclature. *Hepatology*, 78(6):1966–1986, 2023.
4. European Association for the Study of the Liver (EASL), European Association for the Study of Diabetes (EASD) and European Association for the Study of Obesity (EASO). EASL-EASD-EASO Clinical Practice Guidelines on the management of metabolic dysfunction-associated steatotic liver disease (MASLD). *Journal of Hepatology*, 81(3):492–542, 2024. doi:10.1016/j.jhep.2024.04.031.
5. Vilar-Gomez, E., et al. Weight loss through lifestyle modification significantly reduces features of nonalcoholic steatohepatitis. *Gastroenterology*, 149(2):367–378, 2015. doi:10.1053/j.gastro.2015.04.005.
6. Karlas, T., Petroff, D., Sasso, M., Fan, J., Mi, Y., de Lédinghen, V., Kumar, M., et al. Individual patient data meta-analysis of controlled attenuation parameter (CAP) technology for assessing steatosis. *Journal of Hepatology*, 66(5):1022–1030, 2017. doi:10.1016/j.jhep.2016.12.022.
7. National Center for Health Statistics. National Health and Nutrition Examination Survey: Liver Ultrasound Transient Elastography (LUX) data documentation, 2017–March 2020 and August 2021–August 2023. Centers for Disease Control and Prevention, Hyattsville, MD, 2024.
8. National Academies of Sciences, Engineering, and Medicine. Foundational Research Gaps and Future Directions for Digital Twins. The National Academies Press, Washington, DC, 2024. doi:10.17226/26894.
9. Laubenbacher, R., Mehrad, B., Shmulevich, I. and Trayanova, N. Digital twins in medicine. *Nature Computational Science*, 4(3):184–191, 2024. doi:10.1038/s43588-024-00607-6.
10. Katsoulakis, E., Wang, Q., Wu, H., Shahriyari, L., Fletcher, R., Liu, J., Achenie, L., Liu, H., et al. Digital twins for health: a scoping review. *npj Digital Medicine*, 7(1):77, 2024. doi:10.1038/s41746-024-01073-0.
11. Zeevi, D., Korem, T., Zmora, N., Israeli, D., Rothschild, D., Weinberger, A., Ben-Yacov, O., et al. Personalized Nutrition by Prediction of Glycemic Responses. *Cell*, 163(5):1079–1094, 2015. doi:10.1016/j.cell.2015.11.001.

12. Gkouskou, K., Vlastos, I., Karkalousos, P., Chaniotis, D., Sanoudou, D. and Eliopoulos, A. G. The ``Virtual Digital Twins'' Concept in Precision Nutrition. *Advances in Nutrition*, 11(6):1405–1413, 2020. doi:10.1093/advances/nmaa089.

13. Joshi, S., Shamanna, P., Dharmalingam, M., Vadavi, A., Keshavamurthy, A., Shah, L., et al. Digital Twin-Enabled Personalized Nutrition Improves Metabolic Dysfunction-Associated Fatty Liver Disease in Type 2 Diabetes: Results of a 1-Year Randomized Controlled Study. *Endocrine Practice*, 29(12):960–970, 2023. doi:10.1016/j.eprac.2023.08.016.

14. ARTEMIS: AcceleRating the Translation of virtual twins towards a pErsonalised Management of steatotIc liver patientS. Horizon Europe grant 101136299; ClinicalTrials.gov NCT07430501, 2024.

15. Bagler, G. and Goel, M. Computational gastronomy: capturing culinary creativity by making food computable. *npj Systems Biology and Applications*, 10(1):72, 2024. doi:10.1038/s41540-024-00399-5.

16. Batra, D., Diwan, N., Upadhyay, U., Kalra, J. S., Sharma, T., Sharma, A. K., Khanna, D., Marwah, J. S., Kalathil, S., Singh, N., Tuwani, R. and Bagler, G. RecipeDB: a resource for exploring recipes. *Database*, 2020:baaa077, 2020. doi:10.1093/database/baaa077.

17. Garg, N., Sethupathy, A., Tuwani, R., NK, R., Dokania, S., Iyer, A., Gupta, A., et al. and Bagler, G. FlavorDB: a database of flavor molecules. *Nucleic Acids Research*, 46(D1):D1210–D1216, 2018. doi:10.1093/nar/gkx957.

18. Rakhi, N. K., Tuwani, R., Mukherjee, J. and Bagler, G. Data-driven analysis of biomedical literature suggests broad-spectrum benefits of culinary herbs and spices. *PLoS ONE*, 13(5):e0198030, 2018. doi:10.1371/journal.pone.0198030.

19. Sahin, O. and Aytekin-Sahin, G. Open-source multi-objective optimization software for menu planning. *Expert Systems with Applications*, 252:124213, 2024.

20. Bedogni, G., et al. The Fatty Liver Index: a simple and accurate predictor of hepatic steatosis in the general population. *BMC Gastroenterology*, 6:33, 2006.

21. Lee, J., et al. Hepatic steatosis index: a simple screening tool reflecting nonalcoholic fatty liver disease. *Digestive and Liver Disease*, 42(7):503–508, 2010.